



UNIVERSITY OF THE WESTERN CAPE
TUTORIAL 4
AND
MR HEARNS WORKSHEET EXERCISE
WEEK 4 (14 – 18 SEP)

**INFORMATION SYSTEMS
244**

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Group	: 01

Tutorial Answers

1	PL/SQL code
	<pre>SET SERVEROUTPUT ON; ACCEPT emp_id_input NUMBER PROMPT 'Enter the employee id: '; DECLARE iEmp_id HR.employees.employee_id%TYPE := &emp_id_input; sFirst_name HR.employees.first_name%TYPE; sLast_name HR.employees.last_name%TYPE; sJob_id HR.employees.job_id%TYPE; sRow VARCHAR2(200); sBorder VARCHAR2(200); BEGIN SELECT first_name, last_name, job_id INTO sFirst_name, sLast_name, sJob_id FROM HR.employees WHERE employee_id = iEmp_id; sRow := ' ' sLast_name ' ' sFirst_name ' ' sJob_id ' '; sBorder := '+'; FOR i IN 1 .. LENGTH(sRow) - 2 LOOP sBorder := sBorder '-'; END LOOP; sBorder := sBorder '+'; FOR i IN 1 .. LENGTH(sBorder) LOOP DBMS_OUTPUT.PUT(SUBSTR(sBorder, i, 1)); END LOOP; DBMS_OUTPUT.NEW_LINE; DBMS_OUTPUT.PUT_LINE(sRow); FOR i IN 1 .. LENGTH(sBorder) LOOP DBMS_OUTPUT.PUT(SUBSTR(sBorder, i, 1)); END LOOP; DBMS_OUTPUT.NEW_LINE; EXCEPTION WHEN NO_DATA_FOUND THEN DBMS_OUTPUT.PUT_LINE('No employee found with ID ' iEmp_id); END; /</pre>

Output

Script Output x

Task completed in 0.487 seconds

+-----+

| Tobias | Sigal | PU_CLERK |

+-----+

PL/SQL procedure successfully completed.

Enter Value

Enter the employee id:

OK

Cancel

2	PL/SQL code
	<pre> SET SERVEROUTPUT ON; DECLARE CURSOR emp_cursor IS SELECT e.employee_id, e.first_name, e.last_name, e.salary, j.job_title FROM HR.employees e JOIN HR.jobs j ON e.job_id = j.job_id WHERE e.department_id = 60 ORDER BY e.salary DESC; iEmployee_id HR.employees.employee_id%TYPE; sFirst_name HR.employees.first_name%TYPE; sLast_name HR.employees.last_name%TYPE; iSalary HR.employees.salary%TYPE; sJob_title HR.jobs.job_title%TYPE; iCount NUMBER := 0; BEGIN OPEN emp_cursor; LOOP FETCH emp_cursor INTO iEmployee_id, sFirst_name, sLast_name, iSalary, sJob_title; EXIT WHEN emp_cursor%NOTFOUND; iCount := iCount + 1; IF iSalary < 5000 THEN DBMS_OUTPUT.PUT_LINE(sFirst_name ' ' sLast_name ' works as ' sJob_title ' with salary ' iSalary ' (Standard Earner)'); ELSE DBMS_OUTPUT.PUT_LINE(sFirst_name ' ' sLast_name ' works as ' sJob_title ' with salary ' iSalary ' (Higher Earner)'); END IF; END LOOP; CLOSE emp_cursor; END; / </pre>

	Output
	<pre>Alexander Hunold works as Programmer with salary 9000 (Higher Earner) Bruce Ernst works as Programmer with salary 6000 (Higher Earner) David Austin works as Programmer with salary 4800 (Standard Earner) Valli Pataballa works as Programmer with salary 4800 (Standard Earner) Diana Lorentz works as Programmer with salary 4200 (Standard Earner)</pre>   Search    

3 PL/SQL code

```
SET SERVEROUTPUT ON;

CREATE OR REPLACE PROCEDURE dept_salary_report (proc_dept_id IN
HR.departments.department_id%TYPE)
IS
    sDept_name    HR.departments.department_name%TYPE;
    nAvg_salary    NUMBER;
    sBorder        VARCHAR2(60) := RPAD('-', 60, '-');

    CURSOR emp_cursor IS
        SELECT first_name,
               last_name,
               job_id,
               salary
        FROM    HR.employees
        WHERE   department_id = proc_dept_id
        ORDER BY salary DESC, employee_id;

    sTag          VARCHAR2(20);
BEGIN
    SELECT department_name
    INTO    sDept_name
    FROM    HR.departments
    WHERE   department_id = proc_dept_id;

    SELECT AVG(salary)
    INTO    nAvg_salary
    FROM    HR.employees
    WHERE   department_id = proc_dept_id;

    DBMS_OUTPUT.PUT_LINE('Department: ' || sDept_name);
    DBMS_OUTPUT.PUT_LINE('Average Salary: ' || TO_CHAR(nAvg_salary,
                                                         'FML999G999G990D00',
                                                         'NLS_NUMERIC_CHARACTERS=''',
                                                         'NLS_CURRENCY=''R'''));

    DBMS_OUTPUT.PUT_LINE(sBorder);

    FOR emp_rec IN emp_cursor
    LOOP
        IF emp_rec.salary > nAvg_salary THEN
            sTag := '(Above Avg)';
        ELSE
            sTag := '(Below Avg)';
        END IF;

        DBMS_OUTPUT.PUT_LINE(
            RPAD(emp_rec.first_name || ' ' || emp_rec.last_name, 20) ||
            RPAD(emp_rec.job_id, 12) ||
            LPAD(TO_CHAR( emp_rec.salary,
                        'FML999G999G990',
                        'NLS_NUMERIC_CHARACTERS=''',
                        'NLS_CURRENCY=''R'''), 8) || ' ' || sTag);
    END LOOP;

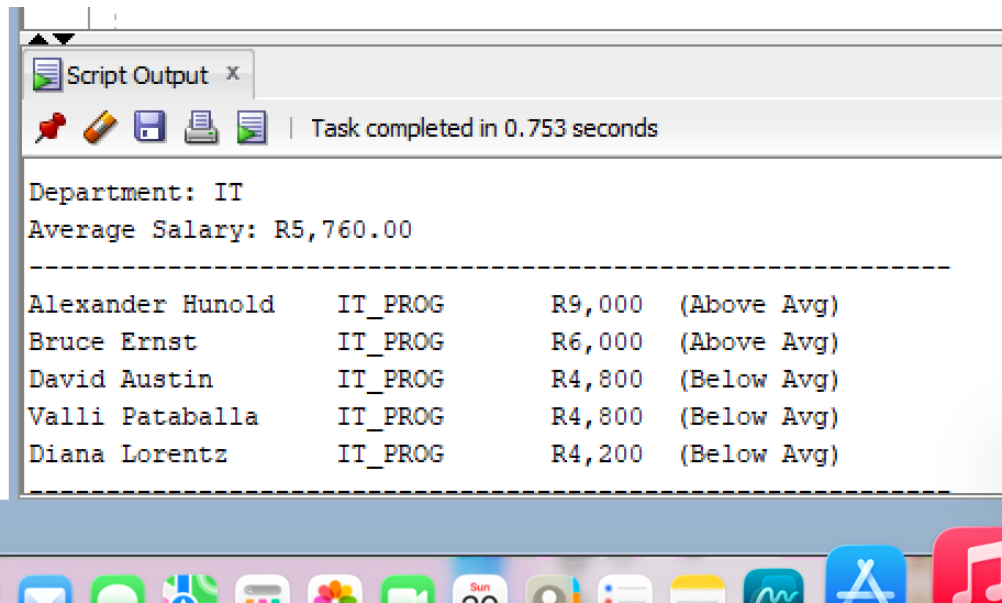
    DBMS_OUTPUT.PUT_LINE(sBorder);
```

```
EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('No department found with ID ' || proc_dept_id || '.');

END dept_salary_report;
/

BEGIN
    dept_salary_report(60);
END;
/
```

Output



MR HEARN'S WORKSHEET EXERCISE ANSWERS

1	PL/SQL code : Exercise 1
	<pre>SET SERVEROUTPUT ON; DECLARE iNumber_of_records NUMBER := &number_of_records; sStart_id HR.employees.employee_id%TYPE := &id; sCurrent_id HR.employees.employee_id%TYPE; sLast_name HR.employees.last_name%TYPE; BEGIN DBMS_OUTPUT.PUT_LINE(iNumber_of_records ' records requested. '); DBMS_OUTPUT.PUT_LINE(' '); FOR i IN 0 .. iNumber_of_records - 1 LOOP sCurrent_id := sStart_id + i; SELECT last_name INTO sLast_name FROM HR.employees WHERE employee_id = sCurrent_id; DBMS_OUTPUT.PUT_LINE(sCurrent_id ' ' sLast_name); END LOOP; EXCEPTION WHEN NO_DATA_FOUND THEN DBMS_OUTPUT.PUT_LINE('No data was found from ID ' sCurrent_id ' '); END; /</pre>

Output
 Task completed in 5.977 seconds <pre>END; 4 records requested. 111 Sciarra 112 Urman 113 Popp 114 Raphaely</pre> perform "Go to Declaration"  Search <pre>5 records requested. No data was found from ID 207. PL/SQL procedure successfully completed.</pre> perform "Go to Declaration"  Search

2 PL/SQL code : Exercise 2

```

SET SERVEROUTPUT ON;
DECLARE
    CURSOR emp_cursor IS
        SELECT e.employee_id,
               e.first_name,
               e.last_name,
               e.salary,
               j.job_title
        FROM   HR.employees e
        JOIN   HR.jobs j ON e.job_id = j.job_id
        WHERE  e.department_id = 60
        ORDER BY e.salary DESC;

    iEmployee_id   HR.employees.employee_id%TYPE;
    sFirst_name    HR.employees.first_name%TYPE;
    sLast_name     HR.employees.last_name%TYPE;
    iSalary        HR.employees.salary%TYPE;
    sJob_title     HR.jobs.job_title%TYPE;

    iCount NUMBER := 0;
BEGIN
    OPEN emp_cursor;
    LOOP
        FETCH emp_cursor
            INTO iEmployee_id,
                sFirst_name,
                sLast_name,
                iSalary,
                sJob_title;
        EXIT WHEN emp_cursor%NOTFOUND;

        iCount := iCount + 1;

        IF iSalary < 5000 THEN
            DBMS_OUTPUT.PUT_LINE(sFirst_name || ' ' || sLast_name ||
                                  ' works as ' || sJob_title || ' with salary ' || iSalary
            ||
                                  ' (Standard Earner)');
        ELSE
            DBMS_OUTPUT.PUT_LINE(sFirst_name || ' ' || sLast_name ||
                                  ' works as ' || sJob_title || ' with salary ' || iSalary
            ||
                                  ' (Higher Earner)');
        END IF;
    END LOOP;
    CLOSE emp_cursor;
END;
/

```

	Output
	Alexander Hunold works as Programmer with salary 9000 (Higher Earner) Bruce Ernst works as Programmer with salary 6000 (Higher Earner) David Austin works as Programmer with salary 4800 (Standard Earner) Valli Pataballa works as Programmer with salary 4800 (Standard Earner) Diana Lorentz works as Programmer with salary 4200 (Standard Earner) 